

DEPARTMENT OF COMPUTER SCIENCE

Gujarat University

MCA – III

Subject: Computer Vision

Practical Assignment – 3

1. Implement Gaussian and Laplacian image pyramids to analyze vehicles of different sizes in a traffic image.
2. Implement an image-pyramid-based approach to detect faces appearing at different scales in a photograph.
3. Implement multi-scale image processing to detect a person appearing at different distances from a surveillance camera.
4. Generate different-resolution versions of a scanned document using Gaussian pyramids and analyze the effect of resolution on character readability.
5. Implement Gaussian derivative filters to detect road boundaries and lane markings in a road image.
6. Implement first-order and second-order Gaussian derivative filters to detect boundaries and defects on a manufactured product.
7. Implement Gabor filters with different orientations and frequencies to identify defective regions in a fabric image.
8. Apply Gabor filters to enhance fingerprint ridge patterns and visualize the resulting texture information.
9. Implement Gabor-filter-based texture analysis to distinguish between smooth and rough or cracked road surfaces.
10. Extract texture features using Gabor filters and classify images of different types of wood.
11. Use Gabor filters to distinguish between healthy and diseased regions of a plant leaf based on texture characteristics.
12. Apply Gabor filters to identify and differentiate surface textures in images of tiles, metal, or other manufactured materials.
13. Implement two-dimensional Discrete Wavelet Transform (DWT) for image compression and reconstruct the image using selected wavelet coefficients.
14. Implement DWT-based image denoising on a noisy medical image and compare the original, noisy, and denoised images.
15. Apply DWT-based denoising to an old or degraded photograph while preserving important image details.
16. Implement DWT-based digital watermarking by embedding a watermark into an image and recovering the embedded watermark.

17. Analyze the quality of two images using their DWT coefficients and compare the amount of detail information present in each image.
18. Implement the Region Growing segmentation technique to segment a selected region from a medical image using suitable seed points.
19. Implement Region Growing to separate water, vegetation, and land regions from a satellite image.
20. Implement edge-based image segmentation using the Canny edge detector to identify road boundaries in a road image.
21. Implement edge-based segmentation to separate a foreground object from its background in a real-world photograph.
22. Implement Graph-Cut-based image segmentation to separate a region of interest, such as an organ, from the background in a medical image.
23. Implement Graph Cut to remove the background from a photograph and extract the required foreground object.
24. Implement Mean-Shift segmentation to divide a natural scene image into meaningful regions based on color and spatial information.
25. Implement Markov Random Field (MRF)-based image segmentation to classify pixels into different regions in a noisy image and compare the result with basic segmentation.
26. Implement texture-based image segmentation using Gabor-filter or DWT-based features to separate regions with different textures, such as grass, road, and building surfaces.
27. Implement an object detection system to detect and count cars, motorcycles, and buses in a traffic image or video.
28. Develop an object detection and recognition system to identify garbage and classify it into categories such as plastic, paper, metal, glass, and organic waste.
29. Develop an object detection and recognition system to identify different products on a store shelf and display their corresponding class labels.
30. Develop a real-time object detection and recognition system for smart-city surveillance that detects predefined objects such as people and vehicles, draws bounding boxes around them, and displays their confidence scores.